

CHAPTER FOUR

RE-EXPRESSING VARIABLES

EXERCISES

Exercises 4.1 – 4.8 involve the effects of linear transformations on the summary statistics of a distribution.

4.1. In this series of exercises, we find the summary statistics of distributions that are linear transformations of original distributions. These exercises do not require the use of SPSS.

- a) If each value in a distribution with mean equal to 5 has been tripled, what is the new mean?
- b) If each value in a distribution with standard deviation equal to 5 has been tripled, what is the new standard deviation?
- c) If each value in a distribution with skewness equal to 1.14 has been tripled, what is the new skewness?
- d) If each value in a distribution with mean equal to 5 has the constant 6 added to it, what is the new mean?
- e) If each value in a distribution with standard deviation equal to 5 has the constant 6 added to it, what is the new standard deviation?
- f) If each value in a distribution with skewness equal to 1.14 has the constant 6 added to it, what is the new skewness?
- g) If each value in a distribution with mean equal to 5 has been multiplied by -2, what is the new mean?
- h) If each value in a distribution with standard deviation equal to 5 has been multiplied by -2, what is the new standard deviation?
- i) If each value in a distribution with skewness equal to 1.14 has been multiplied by -2, what is the new skewness?
- j) If each value in a distribution with mean equal to 5 has had a constant equal to 6 subtracted from it, what is the new mean?
- k) If each value in a distribution with standard deviation equal to 5 has had a constant equal to 6 subtracted from it, what is the new standard deviation?
- l) If each value in a distribution with skewness equal to 1.14 has had a constant equal to 6 subtracted from it, what is the new skewness?

4.2. Consider a situation in which you would like to measure the amount of time it takes children in grades 4 and 6 to complete a task. This exercise does not require SPSS.

- a) The grade 4 teacher found that the average time to complete a task was 20 minutes with a standard deviation of 7 minutes. The grade 6 teacher found that the average time was .25 hours to complete a task with a standard deviation of .09 hours. In which grade were the students faster? In which grade were the students more similar in the amount of time taken to complete

the task?

- b) The grade 4 teacher found that the average time was 20 minutes with a standard deviation of 7 minutes and the grade 6 teacher found that the average time was 22 minutes with a standard deviation of 5 minutes. If the grade 6 teacher had included reading the instructions, which took four minutes, in his calculation of the time to complete the task. after adjusting for this additional time to make the two classes comparable, in which grade were the students faster? In which grade were the students more similar in the amount of time taken to complete the task?

4.3. Assume that you are studying college students and have measured a variable called CREDITS that gives the number of credits taken by a part-time or full-time student during the current semester. For the individuals you are studying, the number of credits taken ranges from 4 to 26, with a mean of 16.26, a standard deviation of 2.40, and a skewness ratio of -12.05. This exercise does not require SPSS.

- a) Assuming that, on average, each class is worth 3 credits, define an SPSS Compute statement to create a variable called CLASSES as a linear transformation of CREDITS that estimates the average number of classes taken during the current semester.
- b) What is the mean of CLASSES?
- c) What is the standard deviation of CLASSES?
- d) What is the shape of the distribution of CLASSES? Explain and support your answer with an appropriate descriptive statistic. Be sure to indicate whether or not the distribution is *severely* skewed.
- e) Why do you suppose the skewness is so severe in the negative direction?

4.4. The variable SCHATTRT in the *NELS* data set gives the average daily attendance rate for the school that the student attends.

- a) What are the mean, median, standard deviation, variance, range, interquartile range, and skewness of SCHATTRT?
- b) Compute a new variable SCHATTPP that expresses the average daily attendance as a decimal instead of as a percentage, i.e., that gives the average daily attendance proportion for the school that the student attends. What is the numeric expression you used?
- c) Based on the summary statistics for SCHATTRT that you generated in part a), find the mean, median, standard deviation, variance, interquartile range, and skewness of SCHATTPP.
- d) Use SPSS to calculate the mean, median, standard deviation, variance, range, interquartile range, and skewness of SCHATTPP.

4.5. The variable EXPINC30 in the *NELS* data set measures the expected annual income at age 30 of students in the eighth grade.

- a) What are the mean, median, standard deviation, variance, range, interquartile range, and skewness of EXPINC30?
 - b) Compute a new variable, EXPCENTS, which gives the expected income at age 30 in terms of pennies, not dollars. What is the numeric expression you used?
 - c) Based on the summary statistics for EXPINC30 that you generated in part a), find the mean, median, standard deviation, variance, interquartile range, and skewness of EXPCENTS.
 - d) Use SPSS to calculate the mean, median, standard deviation, variance, range, interquartile range, and skewness of EXPCENTS.
- 4.6.** The variable COMPUTER in the *NELS* data set indicates whether the student's family owned a computer when the student was in eighth grade. The variable is coded so that 0 = No and 1 = Yes.
- a) Verify that in using this coding, the mean equals the proportion of Yes's, or, in this case, 1's. What is that value?
 - b) Use SPSS to compute the standard deviation of the variable COMPUTER?
 - c) What is the linear transformation to convert COMPUTER into COMP1, which is coded 1 for No and 2 for Yes.
 - d) What is the linear transformation to convert COMPUTER into COMP2, which is coded 0 for Yes and 1 for No.
 - e) Use the rules for the effects of linear transformations on summary statistics to determine the means and standard deviations of the variables COMP1 and COMP2.
- 4.7.** The variable CIGPDAY1 from the *Framingham* data set gives the number of cigarettes smoked per day by the respondents at first examination in 1956. Assume that the values of CIGPDAY1 are then multiplied by negative 2 and then 3 is added to each value to create the variable CIGTRANS.
- a) What are the values of the mean, standard deviation, and skewness ratio of CIGPDAY1?
 - b) What is the value of the mean of CIGTRANS?
 - c) What is the value of the new SD?
 - d) What is the value of the new skewness statistic?
- 4.8.** In the *Framingham* data set, SEX is coded so that 1 = Men and 2 = Women. What is the mean of SEX? Apply appropriate linear transformations to interpret the mean of SEX as a proportion of women.

Exercises 4.9 – 4.17 involve z-scores.

- 4.9.** Consider the variable, number of members in a student's household, given by the *NELS* variable, FAMSIZE (family size).

- a) If a student has a z -score of -1.28, how many members are in her household?
- b) What is the z -score for a student with 6 members in her household?
- c) Using the z -score criteria for outliers, that is, that a score is an outlier if its associated z -score is less than -2 or greater than 2, how many outliers are in this distribution?

4.10. Consider the variable, self-concept in eighth grade, given by the *NELS* variable, SLFCNC08 .

- a) Find the mean and standard deviation of SLFCNC08.
- b) Write down an SPSS compute statement to calculate the z -score distribution for SLFCNC08. Use the SPSS Compute procedure to calculate ZSCORE, the z -score distribution for SLFCNC08.
- c) Use the SPSS Descriptives procedure to verify your calculated z -scores for SLFCNC08. The name given to this variable by SPSS is ZSLFCNC0.
- d) Without computing, what are the means and standard deviations of ZSCORE and ZSLFCNC0?
- e) If a new variable, SLF08P5, were created by adding 5 points to the eighth grade self concept score for each student in the *NELS* data set, what would be the mean and standard deviation of SLF08P5?

4.11. The variable UNITENGL in the *NELS* data set gives the number of years of English taken in high school. The variable UNITMATH gives the number of years of math taken in high school. Advanced course work enabled some of the recorded values of these two variables in the data set to exceed four.

- a) What is the z -score of a student who took 5.5 years of high school English?
- b) How many years of high school English did a student with a z -score of -1.71 take?
- c) Using the z -score criteria for outliers, that is, that a score is an outlier if its associated z -score is less than -2 or greater than 2, how many outliers are in the distribution UNITENGL?
- d) According to cumulative percentages, if a student took 5.5 years of both math and English, relative to his classmates, is that more unusual for English or math?
- e) According to z -scores, if a student took 5.5 years of both math and English, relative to his classmates, is that more unusual for English or math?

4.12. The variables SLFCNC08, SLFCNC10, SLFCNC12 in the *NELS* data set, measure self-concept in eighth, tenth, and twelfth grade, respectively. A self-concept score of 25 is relatively highest in which of these distributions?

- a) Base your answer on cumulative percentages.
- b) Base your answer on z -scores.

4.13. Does an eighth grade self-concept (SLFCNC08) score of 25 represent a higher

level of eighth grade self-concept for males or for females? Use as criteria (a) cumulative percentages and (b) z-scores. Use Split File to create separate frequency distributions for males and female students in the *NELS*.

To get separate frequency distribution tables for males and females, first split the file. Go to **Data, Split File**. Click the circle next to Compare Groups and move GENDER into the box for Groups Based on. Click **OK**. All subsequent commands will be performed separately for males and females. After getting the frequency distributions, turn off the split file by going to **Data, Split File**. Click the circle next to Analyze all cases.

- 4.14. Who scores higher in science achievement in eighth grade (ACHSCI08) relative to his or her gender, a female who scores 58, or a male who scores 63? Use as criteria (a) cumulative percentages and (b) z-scores. Use Split File to create separate frequency distributions for males and female students in the *NELS*.
- 4.15. Determine the number of standard deviations away from the mean a score of 89 is on a test with mean 92 and variance 4.
- 4.16. If Steve's z-score on a test is +1.5, what is his raw score if the mean of the test is 75 and the standard deviation is 10?
- 4.17. Given a unimodal, symmetric distribution, rank from numerically smallest to numerically largest the following values from that distribution:
- $z = +1$ $\bar{X}$ Q_1

Exercises 4.18 – 4.20 involve a mixture of topics about linear transformations and some review topics.

- 4.18. The following questions involve the variable GRADE from the *Learndis* data set.
- a) What are the mean, standard deviation, and skewness ratio of the variable GRADE?
 - b) Assume that students start first grade at age 6 and go up one grade each year. Use a linear transformation to convert GRADE to AGE, the approximate age of the student in that grade.
 - c) Use the information about the distribution GRADE to find the mean age for students in the data set.
 - d) Use the information about the distribution GRADE to find the standard deviation for AGE.
 - e) Use the information about the distribution GRADE to find the skewness ratio for AGE. How would you describe the shape of the AGE distribution?
- 4.19. Compare the distribution of reading comprehension scores (READCOMP) in the *Learndis* data set by whether the student was assigned part time to the resource room for additional instruction (PLACEMENT = 0) or full time in a self-contained

classroom (PLACEMEN = 1).

- a) Is either of the two distributions severely skewed? If so, which one(s) and what is the direction of the skew? Explain and support your answer with appropriate descriptive statistics.
- b) In which type of placement do these students have a higher overall level of reading comprehension? Explain and support your answer with an appropriate descriptive statistic taking into account the varying degrees of skewness in the two distributions. Is there another statistic that suggests an alternative conclusion?
- c) For which type of placement are the reading comprehension scores more consistent? Explain and support your answer with an appropriate descriptive statistic taking into account the varying degrees of skewness in the two distributions. Is there another statistic that suggests an alternative conclusion?
- d) Does the student with the highest reading comprehension score in the data set have a resource room or a self-contained classroom placement? Provide statistical support your answer.
- e) What is the z-score corresponding to a reading comprehension score of 75 for a student with a resource room placement?
- f) What is the reading comprehension score corresponding to a z-score of 2 for a student with a resource room placement?
- g) Would a student with a reading comprehension score of 75 be more unusual (further from the bulk of the scores in standard deviation units) relative to the resource room students or to the self-contained classroom students? Explain and support your answer.
- h) Can a percentile in the distribution of reading comprehension scores for the students with a resource room placement have the value 103? Explain.

4.20. Consider the variable AGE1 in the *Framingham* data set, which gives a respondent's age in 1956, the first year of the study.

- a) How many people in the dataset were 40 years old in 1956?
- b) What percentage of people in the dataset were younger than or equal to 40 in 1956?
- c) Is 40 years old above or below the median in this data set? Explain.
- d) Is 40 years old above or below the mean in this data set? How many standard deviations is 40 years old above or below the mean?
- e) If we consider a score to be an outlier if its associated z-score is less than -2 or greater than 2 , how many outliers are there in the AGE1 distribution?
- f) What is the breakdown of outliers by men and women?
- g) Write the formula for linearly transforming the variable AGE1 to the new variable BIRTHYR, which gives the year in which the person was born.
- h) What is the median of the variable BIRTHYR?
- i) What is the standard deviation of the BIRTHYR?

Exercises 4.21 – 4.23 involve square root and logarithmic transformations.

- 4.21. Consider the four achievement measures in the tenth grade, ACHMAT10, ACHRDG10, ACHHIS10, and ACHSCI10. Determine their skewness and a non-linear transformation (e.g., log, square root) that will be useful in symmetrizing the variable if it is highly skewed.
- 4.22. For each of the following variables from the *NELS* data set, determine whether the variable is skewed, and, if so, whether a log or square root transformation is effective in symmetrizing the variable. In part a) after clicking Transform/Compute to create your new variable(s), click on Paste instead of OK to generate the corresponding Syntax file. Run the commands that appear in the Syntax file to create your new variable(s).
- a) APOFFER
 - b) FAMSIZE
 - c) SCHATTRT
 - d) What would be the effect, if any, of a non-linear transformation on a dichotomous variable, such as CIGARETT?
- 4.23. For each of the following variables from the *Learndis* data set, determine whether the variable is skewed, and, if so, whether a log or square root transformation is effective in symmetrizing the variable.
- a) GRADE
 - b) MATHCOMP
 - c) READCOMP

Exercises 4.24 -4.28 involve recoding and combining variables.

- 4.24. In the *NELS* data set, UNITMATH represents the total number of units (in years) of mathematics taken in high school, while UNITCALC represents the number of units (in years) of calculus taken.
- a) Create the variable UNITMNC to represent the number of units of non-calculus mathematics taken. Generate the Syntax file by clicking **Paste** instead of **OK** and run the commands from the Syntax file to create UNITMNC. What does the Syntax file contain?
 - b) What is the largest number of units of non-calculus math taken by a student in the *NELS* data set? How many student(s) took that amount of non-calculus math? Add to the Syntax file generated by part (a) by clicking Paste (and not OK) after you set up the appropriate analyses from the menu to answer the question of part (b).
- 4.25. There are four variables in the *NELS* data set that describe different types of achievement in twelfth grade: ACHMATH12, ACHRDG12, ACHSCI12, and ACHSLS12.

- a) Create a composite variable, ACHTOT12 that represents the sum of a student's achievement on all four tests.
 - b) Describe the shape of the distribution of ACHTOT12.
 - c) On average, do boys score higher on ACHTOT12 than girls? Support your answer with appropriate descriptive statistics and indicate if different statistics lead to different conclusions.
- 4.26.** The variable APOFFER gives the number of Advanced Placement (AP) courses offered by the school that the student attends.
- a) Create a new variable, APOFFYN, that indicates whether or not the school offers any AP courses, where 0 = No and 1 = Yes. Make sure to record missing data on APOFFER as missing data on APOFFYN.
 - b) How many schools in the *NELS* data set do not offer AP courses?
 - c) How may we interpret the mean of this 0-1 coded dichotomous variable APOFFYN?
- 4.27.** In Exercise 1.15, you created the *Statisticians* data set. Using the variables in that data set, compute the following new variables and answer related questions about them. Generate a single Syntax file for both questions and create the new variables by running the relevant commands in that Syntax file.
- a) Create the variable ALIVE, representing how long they each lived. Who lived the longest?
 - b) Create the variable OLD, that indicates how old they would be now, if they were still living. Who is the "youngest" statistician in the data set?
 - c) What does your Syntax file contain?
- 4.28.** Use the *Framingham* data set to determine if after experiencing a coronary heart disease event (CHD), individuals reduce their weight (as measured by BMI) and cigarette smoking (as measured by CIGDAY). Consider the initial period as period 1 (1956) and the final period as period 3 (1968). The variable ANYCHD4 indicates whether a person in the study experienced a CHD during the study.
- a) Compute two new variables that reflect the difference in BMI and CIGDAY from period 1 to period 3; e.g., $BMIDIFF = BMI3 - BMI1$ and $CIGPDIFF = CIGDAY3 - CIGDAY1$. Looking at the frequency distribution of these two variables, how many individuals lost weight? How many reduced their cigarette smoking?
 - b) What is the change in BMI, on average, from period 1 to period 3 for those who experienced a CHD event during this time period as compared to those who did not experience a CHD event during this time period?
 - c) What is the change in CIGDAY, on average, from period 1 to period 3 for those who experienced a CHD event during this time period as compared to those who did not experience a CHD event during this time period?